

Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with egyptian, roman, gregorian.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Egyptian solar calendar.** 365 days, 12 months of 30 days + 5 epagomenal days. No leap year initially. Synchronised with the Nile flood.
2. **The Roman / Julian / Gregorian.** Julian (45 BCE) — 365.25 days, leap year every 4. Gregorian (1582) — refined to 365.2425, leap years every 4 except centuries unless divisible by 400.
3. **The Indian solar calendar.** *Saura māsa* = time Sun spends in one rāśi. ~30-31 days per month (varies because Sun moves faster near perihelion). Used in Tamil Nadu, Kerala, Bengal — these regions use sidereal solar months even when their festivals are lunar.
4. **The Chinese solar calendar.** 24 *jiéqì* (solar terms) anchored to specific solar longitudes. Spring Equinox, Summer Solstice, etc. are explicit anchor points.
5. **Comparison exercise.** Each student picks 2 traditions and draws a side-by-side year-circle, noting where their respective “new year” falls and what it celebrates.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Cross-Cultural Solar Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).